

Learning and prediction

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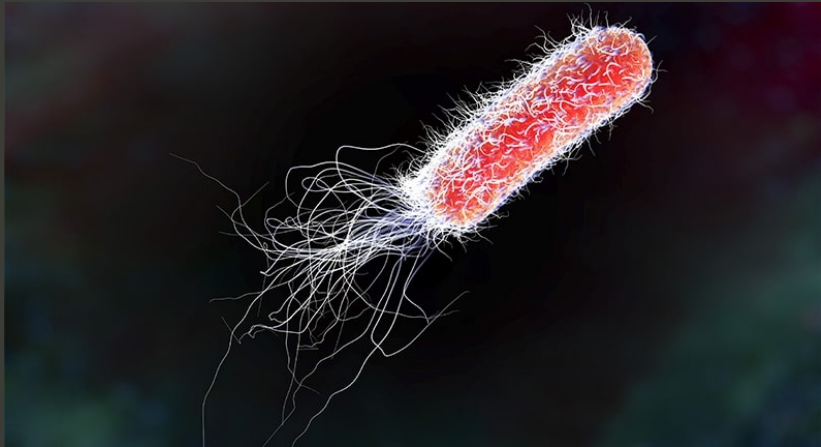
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Learning is the process of gaining knowledge through studying.

From reality to models



From reality to models



From reality to models



$$\vec{f}(\vec{x}, t)$$

$$\left\{ \begin{array}{l} \dot{\vec{y}} = \frac{d\vec{f}(\vec{x}, t)}{d\vec{x}} + \frac{d\vec{f}(\vec{x}, t)}{dt} \\ \vec{x}_0 \end{array} \right.$$

$$y = mX + q$$

From reality to models

$$\vec{f}(\vec{x}, t)$$

$$\begin{cases} \dot{\vec{y}} = \frac{d\vec{f}(\vec{x}, t)}{d\vec{x}} + \frac{d\vec{f}(\vec{x}, t)}{dt} \\ \vec{x}_0 \end{cases}$$

$$y = mX + q$$



```
0100001
1101100
0010000
0110110
1101001
```

+



+



+

```
print("Hello World")
for el in list
```

From reality to models

Model Driven

$$\vec{f}(\vec{x}, t)$$

$$\begin{cases} \dot{\vec{y}} = \frac{d\vec{f}(\vec{x}, t)}{d\vec{x}} + \frac{df(\vec{x}, t)}{dt} \\ \vec{x}_0 \end{cases}$$

$$y = mX + q$$



Data Driven

100011100110001000
 01001001100001001
 00011010011100100
 100011001100011001
 00011011100010100
 011010011011000100
 10111010000111001

Model Vs Data -driven

Model driven

- causal description
- hypotheses driven
- build the MODEL then validate on data

Data driven

- learn the model **from** the **DATA**
- working hypothesis: if the data sampling covers all the possible model states, the DATA implicitly contains the model

From reality to models

Model Driven

$$\vec{f}(\vec{x}, t)$$

$$\begin{cases} \dot{\vec{y}} = \frac{d\vec{f}(\vec{x}, t)}{d\vec{x}} + \frac{d\vec{f}(\vec{x}, t)}{dt} \\ \vec{x}_0 \end{cases}$$

$$y = m \cdot x + q$$



Data Driven

```

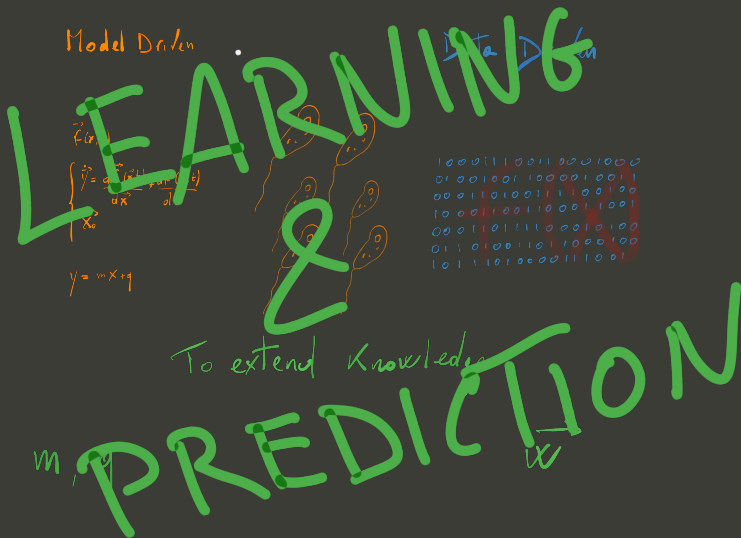
1000111001100010000
010010011000010011
000110100111100100
100011001100011001
00011011100010100
011010011011000100
10111010000111001
  
```

To extend knowledge

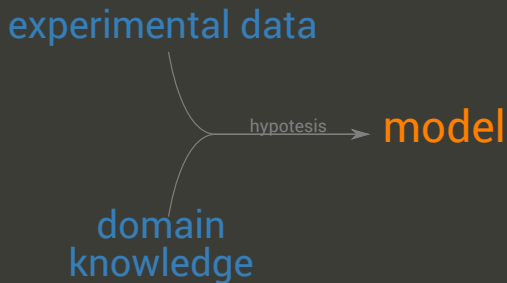
m, q

$\vec{x}$

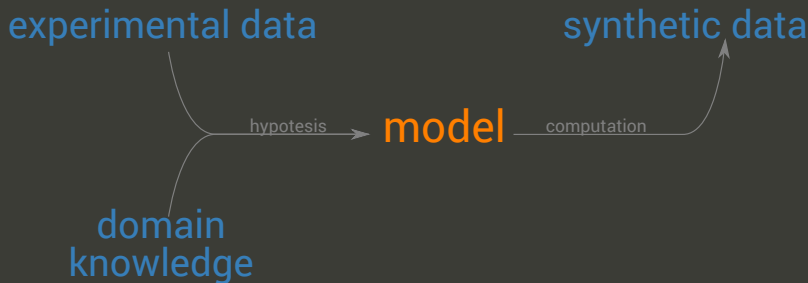
From reality to models



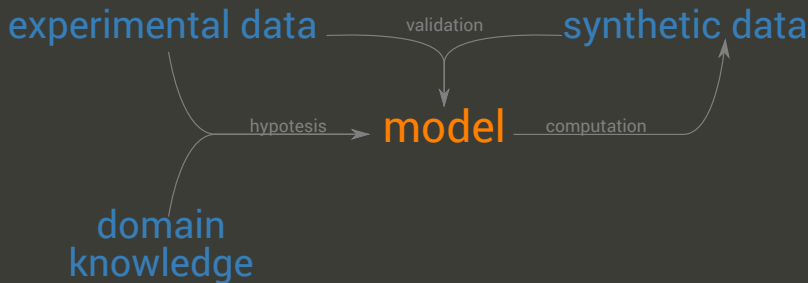
Model-driven



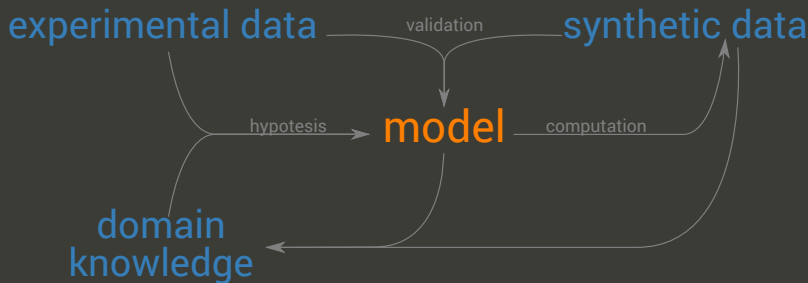
Model-driven



Model-driven



Model-driven



Systems Biology

- Nowadays high throughput technologies give access to a huge amount of data - **quantitative reasoning**
- classic reductionist approach → **system perspective**
- complex system

Systems

- **Complex** system:
 - collection of many simple and identical elements that, interacting, produce a complex behaviour
 - the system functionality emerges from the properties of both the structure and the elements that compose it, and not by each single element
- **Biological** system:
 - collection of many elements with different functionalities that, interacting in a selective and non linear manner, produces a "coherent" behaviour
 - the functionality of the system is a combination of both the structure and the elements that compose it

A systemic approach

- study of all the elements of a biological system, of their individual behaviour and of their interactions
- design and conduction of a reduced number of experiments
- integration, representation and modelling of data that relates to the system in its whole

Systems biology

- the knowledge of the constituents elements of a biological system (genes, proteins, metabolites, ...) is necessary but not sufficient to understand the system in its entirety
- Description and analysis of a biological system:
 - system structure
 - system dynamics
 - control methods
 - design and alteration methods
- Tools for the description and the analysis: collection of principles and methodologies to link the molecular behaviour to the characteristics and the functionalities of the system

Today quantitative study of the biological systems

Today the study of a biological system can be performed through:

- system perturbation at the biological, genetic, chemical or physical level
- the monitoring of genes expression, proteins, metabolites and signaling pathways
- data integration
- management and sharing of data (databases)
- the design and sharing of mathematical models
- performing analyses and validations on the mathematical models